

Waterproof body feathers keep the bird dry.

Underneath its body feathers, the bird wears a warm vest of fluffy down feathers.

This rosella has about 4,000 feathers.

Adult birds lose old feathers and grow new ones. This is called molting.

Primary feathers can twist like propellers to power the bird through the air.

These secondary flight feathers help lift the bird up in the air.

Hooked up
Each flight feather has thousands of fine strands, called barbules. These hook around each other and hold the feather in shape, even in very windy weather.

Tail feathers are used for steering and stopping.

Quill

Vane

Barbule

Barb

Strong shaft

Flapping flight

As this pigeon flaps its wings down, its feathers close so that they can push against the air. On the way up, the feathers open to let the air slip through.

Wash and brush

Birds keep their feathers clean and tidy by preening. They nibble each feather to zip the barbules back together and to get rid of insects. Most birds also waterproof their feathers by rubbing oil from a preen gland which is just above their tail.